

Computer Science and Artificial Intelligence Laboratory (CSAIL) is a research institute at the Massachusetts Institute of Technology (MIT) formed by the 2003 merger of the Laboratory for Computer Science (LCS) and the Artificial Intelligence Laboratory (AI Lab). Housed within the Ray and Maria Stata Center, CSAIL is the largest on-campus laboratory as measured by research scope and membership. It is part of the Schwarzman College of Computing but is also overseen by the MIT Vice President of Research.

Research activities

CSAIL's research activities are organized around a number of semi-autonomous research groups, each of which is headed by one or more professors or research scientists. These groups are divided up into seven general areas of research:

Artificial intelligence

Computational biology

Graphics and vision

Language and learning

Theory of computation

Robotics

Systems (includes computer architecture, databases, distributed systems, networks and networked systems, operating systems, programming methodology, and software engineering, among others)

History

Computing Research at MIT began with Vannevar Bush's research into a differential analyzer and Claude Shannon's electronic Boolean algebra in the 1930s, the wartime MIT Radiation Laboratory, the post-war Project Whirlwind and the Research Laboratory of Electronics (RLE),

and MIT Lincoln Laboratory's SAGE in the early 1950s. At MIT, research in the field of artificial intelligence began in the late 1950s.

Project MAC

On July 1, 1963, Project MAC was launched with a \$2 million grant from the Defense Advanced Research Projects Agency (DARPA). According to the MIT archives:

They chose the project name MAC in 1963 because it is an acronym for several significant phrases describing the project and its goals. "Machine-Aided Cognition" was the broad objective. A "Multiple-Access Computer" was the principal tool. "Men and Computers" were the essential partners in the expected symbiosis, a term used by J. C. R. Licklider in his 1960 paper entitled "Man-Computer Symbiosis."

Project MAC's first director was Robert Fano of RLE. Fano called MAC a "project" rather than a "laboratory" for reasons of internal MIT politics – if MAC had been called a laboratory, it would have been more difficult to raid other MIT departments for research staff. The DARPA program manager was J. C. R. Licklider, who had previously been at RLE, and who later succeeded Fano as director of Project MAC.

Project MAC became famous for groundbreaking research in operating systems, artificial intelligence, and the theory of computation. Its contemporaries included Project Genie at Berkeley, the Stanford Artificial Intelligence Laboratory, and (somewhat later) University of Southern California's (USC's) Information Sciences Institute.

An "AI Group" including Marvin Minsky (the director), John McCarthy (inventor of Lisp), and a community of computer programmers were incorporated into Project MAC. They were interested principally in the problems of vision, mechanical motion and manipulation, and language, which they view as the keys to more intelligent machines. In the 1960s and 1970s the AI Group developed a time-sharing operating system called Incompatible Timesharing

System (ITS) which ran on PDP-6 and later PDP-10 computers.

The early Project MAC community included Fano, Minsky, Licklider, Fernando J. Corbató, and a community of computer programmers and enthusiasts among others who were inspired by John McCarthy. They envisioned the creation of a computer utility whose computational power would be as reliable as an electric utility. To this end, Corbató brought the first computer time-sharing system, Compatible Time-Sharing System (CTSS), with him from the MIT Computation Center, using the DARPA funding to purchase an IBM 7094 for research use.

CTSS was described in an article on time sharing in Scientific American's 1966 special issue on "Information". It had approximately 100 TTY terminals, mostly on campus but with a few in private homes. About 30 users could be logged in at the same time. The project enlisted students in various classes to use the terminals simultaneously in problem solving, simulations, and multi-terminal communications as tests for the multi-access computing software being developed.

One of the early focuses of Project MAC was the development of a successor to CTSS, Multics, which became the first high availability computer system, and a testing ground for the concept of a security kernel. It was initially developed as a part of an industry consortium including General Electric, Bell Laboratories. Following Bell's departure from the project in 1969 and GE's exit from the computer industry in 1970, Project MAC continued to develop Multics throughout the 1970s with Honeywell.

AI Lab and the Laboratory for Computer Science

In the late 1960s, Minsky's artificial intelligence group sought more space, and was unable to get satisfaction from project director Licklider. Minsky found that although Project MAC as a single entity could not get the additional space he wanted, he could split off to form his own laboratory and then be entitled to more office space. As a result, the MIT AI Lab was formed in 1970, and many of Minsky's AI colleagues left Project MAC to join him in the new laboratory.

Programmers such as Richard Stallman, who used TECO to develop EMACS, flourished in the AI Lab during this time. The AI Lab invented Lisp machines, which were commercialized by Symbolics and Lisp Machines Inc. in the 1980s.

Project MAC itself was officially re-named the Laboratory for Computer Science (LCS) in 1976. Participants continued their research into operating systems, programming languages, distributed systems, security kernels, and the theory of computation.

Two professors, Hal Abelson and Gerald Jay Sussman, formed a smaller group positioned between the two. Officially called the MIT Project on Mathematics and Computation (creating a backronym for the discontinued designation "Project MAC"), it was given the playful nickname of "Switzerland" for its ostensible neutrality.

CSAIL

On the fortieth anniversary of Project MAC's establishment, July 1, 2003, LCS was merged with the AI Lab to form the MIT Computer Science and Artificial Intelligence Laboratory, or CSAIL. This merger created the largest laboratory (over 600 personnel) on the MIT campus.

In 2018, CSAIL launched a five-year collaboration program with iFlytek, a company sanctioned the following year for allegedly using its technology for surveillance and human rights abuses in Xinjiang. In October 2019, MIT announced that it would review its partnerships with sanctioned firms such as iFlyTek and SenseTime. In April 2020, the agreement with iFlyTek was terminated.

CSAIL moved from the School of Engineering to the newly formed Schwarzman College of Computing by February 2020.

Offices

From 1963 to 2004, Project MAC, LCS, the AI Lab, and CSAIL had their offices at 545 Technology Square, taking over more and more floors of the building over the years. In 2004,

CSAIL moved to the new Ray and Maria Stata Center, which was built specifically to house it and other departments.

Outreach activities

The IMARA (from Swahili word for "power") group sponsors a variety of outreach programs that bridge the global digital divide. Its aim is to find and implement long-term, sustainable solutions which will increase the availability of educational technology and resources to domestic and international communities. These projects are run under the aegis of CSAIL and staffed by MIT volunteers who give training, install and donate computer setups in greater Boston, Massachusetts, Kenya, Native American Indian tribal reservations in the American Southwest such as the Navajo Nation, the Middle East, and Fiji Islands. The CommuniTech project strives to empower under-served communities through sustainable technology and education and does this through the MIT Used Computer Factory (UCF), providing refurbished computers to under-served families, and through the Families Accessing Computer Technology (FACT) classes, it trains those families to become familiar and comfortable with computer technology.

Notable researchers

(Including members and alumni of CSAIL's predecessor laboratories)

MacArthur Fellows Tim Berners-Lee, Erik Demaine, Dina Katabi, Daniela L. Rus, Regina Barzilay, Peter Shor, Richard Stallman, and Joshua Tenenbaum

Turing Award recipients Leonard M. Adleman, Fernando J. Corbató, Shafi Goldwasser, Butler W. Lampson, John McCarthy, Silvio Micali, Marvin Minsky, Ronald L. Rivest, Adi Shamir, Barbara Liskov, and Michael Stonebraker

IJCAI Computers and Thought Award recipients Terry Winograd, Patrick Winston, David Marr,

Gerald Jay Sussman, Rodney Brooks

Rolf Nevanlinna Prize recipients Madhu Sudan, Peter Shor, Constantinos Daskalakis

Gödel Prize recipients Shafi Goldwasser (two-time recipient), Silvio Micali, Maurice Herlihy, Charles Rackoff, Johan Håstad, Peter Shor, and Madhu Sudan

Grace Murray Hopper Award recipients Robert Metcalfe, Shafi Goldwasser, Guy L. Steele, Jr., Richard Stallman, and W. Daniel Hillis

Textbook authors Harold Abelson and Gerald Jay Sussman, Richard Stallman, Thomas H. Cormen, Charles E. Leiserson, Patrick Winston, Ronald L. Rivest, Barbara Liskov, John Guttag, Jerome H. Saltzer, Frans Kaashoek, Clifford Stein, and Nancy Lynch

David D. Clark, former chief protocol architect for the Internet; co-author with Jerome H. Saltzer (also a CSAIL member) and David P. Reed of the influential paper "End-to-End Arguments in Systems Design"

Eric Grimson, expert on computer vision and its applications to medicine, appointed Chancellor of MIT March 2011

Bob Frankston, co-developer of VisiCalc, the first computer spreadsheet

Seymour Papert, inventor of the Logo programming language

Joseph Weizenbaum, creator of the ELIZA computer-simulated therapist

Notable alumni

Robert Metcalfe, who later invented Ethernet at Xerox PARC and later founded 3Com

Marc Raibert, who created the robot company Boston Dynamics

Drew Houston, co-founder of Dropbox

Colin Angle and Helen Greiner who, with previous CSAIL director Rodney Brooks, founded iRobot

Jeremy Wertheimer, who developed ITA Software used by travel websites like Kayak and Orbitz

Max Krohn, co-founder of OkCupid

Directors

Directors of Project MAC

Robert Fano, 1963–1968

J. C. R. Licklider, 1968–1971

Edward Fredkin, 1971–1974

Michael Dertouzos, 1974–1975

Directors of the Artificial Intelligence Laboratory

Marvin Minsky, 1970–1972

Patrick Winston, 1972–1997

Rodney Brooks, 1997–2003

Directors of the Laboratory for Computer Science

Michael Dertouzos, 1975–2001

Victor Zue, 2001–2003

Directors of CSAIL

Rodney Brooks, 2003–2007

Victor Zue, 2007–2011

Anant Agarwal, 2011–2012

Daniela L. Rus, 2012–

CSAIL Alliances

CSAIL Alliances is the industry connection arm of MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL). CSAIL Alliances offers companies programs to connect with the research, faculty, students, and startups of CSAIL by providing organizations with opportunities to learn about the research, engage with students, explore collaborations with

researchers, and join research initiatives such as FinTech at CSAIL, MIT Future of Data, and Machine Learning Applications.

See also

References

Further reading

"A Marriage of Convenience: The Founding of the MIT Artificial Intelligence Laboratory" (PDF). Archived from the original (PDF) on May 14, 2011. Retrieved December 30, 2007., Chious et al. — includes important information on the Incompatible Timesharing System

Weizenbaum. Rebel at Work: a documentary film with and about Joseph Weizenbaum

Garfinkel, Simson (1999). Abelson, Hall (ed.). Architects of the Information Society: Thirty-Five Years of the Laboratory for Computer Science at MIT. Cambridge, Massachusetts: MIT Press. ISBN 0-262-07196-7.

External links

Official website of CSAIL, successor of the AI Lab